Predicting 24-Month Mortgage Prepayment from Origination Data

Logistic Regression, SVM, Gaussian Discriminant Analysis, and Feed-Forward Neural Network

Authors: Not available in the current repository

1. Abstract

We predict full mortgage prepayment within 24 months using a 50,000-loan Freddie Mac 2019 sample (49,111 retained). Four classifier families share a stratified 20% hold-out set (9,823 loans). Most variants achieve 0.629–0.657 accuracy versus a 0.524 majority-class baseline. The neural network leads in accuracy (0.657), precision (0.676), and ROC-AUC (0.709), but differences between leading models are small.

2. Problem, Dataset and Preprocessing

The target is 1 if a loan prepays in full within 24 months, otherwise 0. It is derived from performance records using Zero Balance Code 01 at loan age ≤24 months; predictors use origination data only. Performance columns are excluded to prevent leakage: current_UPB is exactly zero on every prepayment row and would reveal the label.

After removing 889 loans with repeated loan ages, 49,111 remained. Freddie Mac unavailable codes (9999/999/99) became missing; training-set medians imputed 47 credit-score/DTI values. Fourteen raw features became 20 after encoding. Nine numeric features were standardised; first-time-buyer stayed binary; occupancy, channel, property type and purpose were one-hot encoded. CLTV, MSA, ZIP3, state, seller, first-payment date and constant columns were excluded.
A stratified 80/20 split (random_state=42) preceded preprocessing: 39,288 training and 9,823 test loans. Positive class share was 52.4% (25,749/49,111); no resampling or class weighting was used.

3. Methods

Logistic Regression: L2, lbfgs, threshold 0.5; C selected by 5-fold training-only CV over 0.01–10 (C=1; CV AUC=0.693). SVM: RBF (C=1), polynomial (degree 3, C=1), sigmoid (C=1), scale gamma; settings not tuned. GDA: LDA with shared covariance; QDA reg_param=0.1. Neural network: 20–32–16–1, ReLU hidden layers, sigmoid output (1,217 parameters), binary cross-entropy, Adam (0.001), batch 64, early stopping on 15% validation split; stopped after 20 epochs. Test data was not used for tuning.

4. Test Results

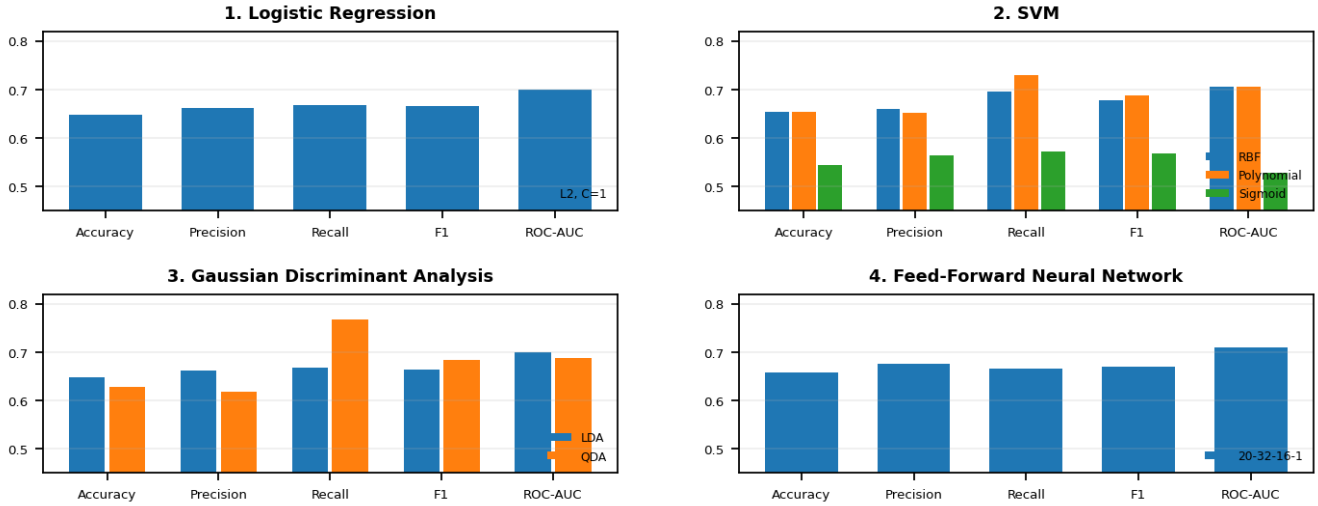
Table 1. Test metrics (n=9,823; positive class = prepayment).

Family	Variant	Acc.	Prec.	Recall	F1	AUC
Logistic Reg.	L2, C=1	.6474	.6621	.6685	.6653	.7004
SVM	RBF	.6534	.6608	.6963	.6781	.7065
SVM	Polynomial	.6540	.6518	.7299	.6887	.7068
SVM	Sigmoid	.5448	.5652	.5715	.5683	.5293
GDA	LDA	.6472	.6622	.6676	.6649	.7003
GDA	QDA	.6289	.6175	.7680	.6846	.6877
Neural Net	20-32-16-1	.6573	.6756	.6664	.6710	.7094

The neural network leads in accuracy, precision and AUC; QDA has the highest recall (0.768), while polynomial SVM has the highest F1 (0.689). No model dominates every metric.

5. Four Model-Family Plots

Test accuracy, precision, recall, F1 and ROC-AUC for each family; values are taken from Table 1.



6. Discussion

Five of seven variants cluster at accuracy 0.647–0.657 and AUC 0.700–0.709. Logistic Regression and LDA are nearly identical. The neural network exceeds Logistic Regression by about 0.010 accuracy and 0.009 AUC; with one split and no confidence intervals, this is modest evidence. QDA predicts more loans as prepaid, reducing false negatives but increasing false positives. Polynomial SVM also trades precision for recall. The sigmoid SVM is near chance (AUC 0.529); this only shows its untuned setting performed poorly.

Prepayment rates vary nonlinearly across interest-rate bands (0.402 → 0.569 → 0.456), yet flexible models improve only marginally, suggesting limited signal. Logistic Regression is more interpretable (loan amount +0.64, owner-occupied +0.63); the co-op coefficient (−1.86) rests on few loans. Train/test accuracy and AUC are similar for Logistic Regression (0.6445/0.6474; 0.6941/0.7004) and the neural network (0.6588/0.6573; 0.7158/0.7094).

7. Limitations, Conclusion and References

Limitations: one 2019 sample; origination-only features; one split/seed; fixed 0.5 threshold; no confidence intervals; correlated features (max |r|=0.82); GDA's Gaussian assumption despite 11 binary columns; only Logistic Regression cross-validated; SVM and NN not tuned. Each SVM took about 400–430 s.

Conclusion: origination features predict above the 0.524 baseline but only moderately (accuracy ≈0.65; AUC ≈0.70). The neural network leads on three metrics, but differences are around one percentage point. Model choice should balance interpretability and the costs of false positives versus false negatives.

References: (1) Freddie Mac Single-Family Loan-Level Dataset, 2019 sample. (2) Reference paper cited in project; title/authors/DOI unavailable in repository. (3) scikit-learn, TensorFlow/Keras, pandas, NumPy, Matplotlib.